

## *Combining Two Random Variables*

Example: Let X and Y be two independent random variables with the following probability distribution.

|      |     |     |     |
|------|-----|-----|-----|
| X    | 1   | 2   | 3   |
| P(X) | 0.3 | 0.2 | 0.5 |

|      |     |     |
|------|-----|-----|
| Y    | 5   | 10  |
| P(Y) | 0.4 | 0.6 |

If we let  $S = X + Y$ , create the probability distribution below:

Now find the following:  $E(S) =$ \_\_\_\_\_  $\sigma_S =$ \_\_\_\_\_

$E(X) =$ \_\_\_\_\_  $\sigma_X =$ \_\_\_\_\_

$E(Y) =$ \_\_\_\_\_  $\sigma_Y =$ \_\_\_\_\_

There is a shortcut for finding means and standard deviations of combined random variables WITHOUT having to create a new distribution!

If X and Y are two independent random variables:

$E(X + Y) =$ \_\_\_\_\_  $E(X - Y) =$ \_\_\_\_\_

$\sigma_{X+Y} =$ \_\_\_\_\_  $\sigma_{X-Y} =$ \_\_\_\_\_

Check the problem above:

# Linear Transformations of Random Variables

Example: The head folks at Google decided to try a random salary assignment for their workers this year! They have the following probability distribution, where  $X$  = salary.

|      |        |        |        |        |
|------|--------|--------|--------|--------|
| X    | 50,000 | 60,000 | 70,000 | 80,000 |
| P(X) | 0.2    | 0.3    | 0.4    | 0.1    |

1) What is the expected salary for a Google employee? What is the standard deviation?

2) Google decides to give EVERYONE a \$500 bonus. Create a new probability distribution and find the expected value and standard deviation. Notice anything?

3) Google now decides that it will pay cash at the end of the year for this salary. They will give out \$100 bills. Let  $X$  = number of \$100 bills they will give out. Create a new probability distribution (use the bonus distribution) and find the expected value and standard deviation. Notice anything?

If  $X$  is a random variable and  $a$  and  $b$  are both constants:

$$E(a + bX) = \underline{\hspace{2cm}}$$

$$\sigma_{a+bX} = \underline{\hspace{2cm}}$$

Example: Depending on the attendance of a minor league baseball team, the number of hot dogs and sodas sold at a game is given by the following table.

|               |      |     |     |       |       |
|---------------|------|-----|-----|-------|-------|
| Hot Dogs Sold | 100  | 200 | 500 | 1,000 | 2,500 |
| P(H)          | 0.05 | 0.1 | 0.3 | 0.35  | 0.2   |

|            |     |      |      |     |       |
|------------|-----|------|------|-----|-------|
| Sodas Sold | 100 | 200  | 400  | 800 | 1,500 |
| P(S)       | 0.1 | 0.15 | 0.25 | 0.3 | 0.2   |

If the profit on a hot dog is \$1.50 and the profit on a soda is \$0.75, find the average profit on hot dogs and sodas per game. What is the standard deviation of this profit?

Example: Suppose two pro-bowlers Larry (L) and Sam (S) have the following distribution of scores:

$$\mu_L = 239$$

$$\sigma_L = 10$$

$$\mu_S = 244$$

$$\sigma_S = 25$$

Find each of the following:

a)  $E(L + S)$

b)  $\mu_{S-L}$

c)  $\sigma_{L+S}$

d)  $3\sigma_L$

e)  $\sigma_{L-S}$

f)  $E(4 + 3L)$